Business Statistics — Open-Book Index

Consolidated index of Lectures 1—16 (498 pp., Chapters 1—12 + Nonparametric Methods) — page references to the source deck for the exam.

Chapter Map *(pp. 1—498)*

Chapter 1: Defining and Collecting Data *(pp. 14—38)*

Measurement scales (p.23-27)	Nominal (categories only, e.g. Gender, Industry) → Ordinal (ranked, e.g. Child/Adult/Senior; Taste ratings) → Interval (measured, arbitrary zero, e.g. Temperature °F) → Ratio (measured, TRUE zero — allows ratio comparisons, e.g. Weight, Sales, Income, Delivery time).
Analysis technique depends on scale (p.28)	Choice of statistical technique (charts, summary measures, tests) is dictated by whether the variable is Nominal/Ordinal/Interval/Ratio.
Sampling methods & survey errors (p.15)	Covered together with Ch.7 (Sampling Distributions) — see Chapter 7 below.

Chapter 2: Organizing & Visualizing Variables *(pp. 29—56)*

One-variable data (p.31, 50)	Single-variable summarization — tables & charts for one categorical or numeric variable.
Cross-sectional data (p.32-33, 45-46)	Data on many subjects at one point in time; grouped into class intervals for large numeric datasets.
Two categorical variables (p.48)	Contingency/cross-tab tables; side-by-side bar charts.
Two numeric variables (p.49)	Scatter plots to visualize relationship between two numeric variables.
Charts (p.34-37, 53-54)	Tree map & Heat map (categorical, hierarchical); Bar/Column chart (categories) vs. Histogram (numeric, continuous class intervals — no gaps between bars); Frequency polygon (line joining histogram midpoints); Time-series / Line chart(s) for trend over time; chart creation in MS Excel.
Analysing a Histogram (p.37, 46)	Look at shape (symmetric/skewed), central location, spread, and presence of gaps/outliers/multiple peaks.

Chapter 3: Numerical Descriptive Measures *(pp. 57—136, 429—433)*

Central Tendency *(p. 61-67, 77)*

Mean	Sum of values / number of observations = $(1/N)\cdot\Sigma x_i$. For grouped data: weighted mean using class midpoints and frequencies (p.77).

Median	Middle value of sorted data (50th percentile / 50-50 split); use when data has outliers or is skewed.
Mode	Most frequently occurring value; use for categorical/nominal data or to find the most common value.

Variation *(p. 80-96)*

Range	Maximum − Minimum. Shortcoming: uses only 2 observations, ignores the rest of the distribution (p.86-87).
Variance (σ^2)	Average of squared deviations from the mean: $\Sigma(x - \text{mean})^2 / N$. Deviations are squared because raw errors from the mean sum to zero (p.88-89).
Standard Deviation (σ)	√Variance — brings the measure back to the original units. Marked in the deck as "the most important formula of the course" (p.85, 88-89).
Coefficient of Variation (CoV)	(Standard Deviation / Mean) × 100. Used to compare variability across datasets with different units or different means (e.g. comparing 200m sprint times vs. Javelin throw distances) — Range/SD alone can't be used when units differ (p.95-96).

Shape *(p. 98-114)*

Skewness	Measures asymmetry. Left tail → negatively skewed; right tail → positively skewed; symmetric → zero skew (p.100-101).
Kurtosis	Measures peakedness relative to the Normal distribution. Positive kurtosis = more peaked than Normal; negative = flatter (p.113).

Exploring Numerical Data *(p. 115—124)*

Quartiles / 5-Number Summary	Sorted data split into 4 equal 25% parts: Q1, Q2 (=Median), Q3. 5-Number Summary = Minimum, Q1, Q2, Q3, Maximum (p.115-116).
Box plot	Visualizes the 5-number summary — box spans Q1 to Q3, line at median, whiskers to min/max (p.117-118).
Z-score	(Observed value − Mean) / Standard deviation — expresses a single observation's distance from the mean in SD units (p.119-121).
Outliers	Typically flagged using Z-score thresholds (e.g. $\lvert Z\rvert > 3$) or the 1.5×IQR rule from the box plot (p.122-124).

Covariance & Correlation *(p. 416—433 (cross-ref Ch.12))*

Covariance	Measures the joint variability of two numeric variables: $\Sigma(x-\bar{x})(y-\bar{y})/(n-1)$ — sign shows direction, but magnitude depends on units (p.416-419).
Correlation coefficient, r	Standardized covariance (range −1 to +1); measures strength & direction of the LINEAR relationship between two numeric variables. r near ±1 → strong relationship, useful for prediction; r near 0 → weak/no linear relationship (p.419-424, 432).

Chapter 4: Basic Probability *(pp. 137—201)*

Sources of probability (p.158-166)	Empirical (relative frequency from observed data); Subjective (personal judgment/belief); A priori / Classical (equally likely outcomes, e.g. dice, cards) — shortcoming: assumes equal likelihood, which often doesn't hold in real business situations.
Marginal, Joint & Conditional probability (p.167-180)	Marginal: P(A) — probability of a single event alone; P(A) = P(A and B) + P(A and Not B). Joint: P(A and B) — both events together. Conditional: P(A|B) — probability of A given B has occurred.
Independent vs. Dependent events (p.181)	Independent: P(A|B) = P(A) — knowledge of B doesn't change the probability of A (e.g. P(King)=P(King|Red)=8%). Dependent: P(A|B) ≠ P(A) (e.g. P(King)=8% but P(King|Picture card)=33%).
Bayes' Theorem (p.182-190)	Revises a prior probability using new (conditional) information: P(A|B) = [P(B|A)·P(A)] / P(B), where P(B) is expanded via the joint probabilities of B with each possible A outcome (e.g. P(Pizza|Frooti) = P(Pizza∩Frooti) / [P(Pizza∩Frooti)+P(Burger∩Frooti)]).

Chapter 5: Discrete Probability Distributions *(pp. 202—257)*

Discrete probability distribution (p.204-218)	Lists all possible values of a discrete random variable with their probabilities. Expected Value (Mean) = $\Sigma[x \cdot P(x)]$ — used e.g. for product warranty cost estimation.
Binomial distribution (p.220-238)	Applies when: (1) only two outcomes per trial — Success/Failure; (2) probability of success p is constant across trials; (3) trials are independent; (4) fixed number of trials, n. P(x successes) computed via the binomial formula using n, p, and x; usable in MS Excel via BINOM.DIST.
Poisson distribution (p.239-243, 254)	Models the number of occurrences of an event over a fixed interval (time/space), used when events are rare/random and independent, e.g. no. of defects, accidents, potholes per km.
Mean & Variance (p.244-245)	Binomial: Mean = $n \cdot p$, Variance = $n \cdot p \cdot (1-p)$. Poisson: Mean = λ (given/known), Variance = λ (Mean = Variance is a key Poisson property).

Chapter 6: The Normal Distribution *(pp. 258—276)*

Normal distribution properties (p.260-262)	Continuous, symmetric, bell-shaped. Total area under the curve = 1 (100% probability). Area from $-\infty$ to Mean (μ) = 50%. Many real-world/business variables approximate Normal, and some discrete distributions (Binomial, Poisson) can be approximated by it for large samples.
Standardized Normal (Z) distribution (p.263-269)	Any Normal variable converted to Z-scores (Mean=0, SD=1) using $Z = (X - \mu)/\sigma$; probabilities then read off the Z-table or via MS Excel (NORM.S.DIST / NORM.DIST).

Chapter 7: Sampling Distributions *(pp. 282—300 (+ Ch.1 sampling methods, p.15, 283))*

Census vs. Sampling; sampling methods (p.285-287)	Census = studying the entire population; Sampling = studying a representative subset. Sampling methods and types of survey errors are covered together with Ch.1.
Parameters vs. Statistics; Estimation (p.288-289)	Parameter = a numeric measure describing the population (e.g. μ, σ, p); Statistic = the corresponding measure computed from a sample (e.g. $\bar{x}$, s, $\bar{p}$), used to estimate the parameter.
Sampling distributions (p.291-293)	The probability distribution of a sample statistic (e.g. of $\bar{x}$) computed over all possible samples of a given size from the population.
Central Limit Theorem, CLT (p.294)	If the population is Normally distributed, the sampling distribution of the mean is also Normal. For sufficiently large sample size (n), the sampling distribution of $\bar{x}$ approaches Normal regardless of the population's shape.
Sampling distribution of the proportion (p.298-299)	Distribution of the sample proportion $\bar{p}$ across repeated samples — also approaches Normal for large n, used as the basis for CI/hypothesis tests on proportions.

Chapter 8: Confidence Interval Estimation *(pp. 304—326)*

Confidence level & t-distribution (p.307-310)	Confidence level = probability that the interval contains the true parameter (e.g. 95%). t (Student) distribution used instead of Z when population σ is unknown — flatter/wider tails than Normal, defined by degrees of freedom (df = n − 1).
CI for the mean (p.311-314)	Point estimate: $\mu \approx \bar{x}$. Interval: $\bar{x} \pm$ (t or z value) × (σ or s)/$\sqrt{n}$.
CI for the proportion (p.315-318)	Point estimate: p ≈ $\bar{p}$ (sample proportion). Interval: $\bar{p}$ ± z value × Standard Error of $\bar{p}$.
Determining sample size (p.319-325)	Rearranging the CI formula for n: for the mean, $n = (z \cdot \sigma/E)^2$ where E is the desired margin of error; analogous formula for the proportion using $\bar{p}(1-\bar{p})$.

Chapter 9: Hypothesis Testing — One-Sample Tests *(pp. 330—356)*

Hypothesis testing logic (p.333-334)	Tests a claim about a population parameter (e.g. average tyre life μ = 100K km) using sample evidence; decision is to Reject or Do-Not-Reject the claim based on whether the sample statistic falls in a critical (rejection) region.
Null (H0) & Alternate (H1) Hypothesis (p.335)	H0: the status-quo/claimed value (e.g. μ = 100K); H1: what is concluded if H0 is rejected (e.g. $\mu \neq$ 100K).
Type-I & Type-II Errors (p.336-337)	Type-I error: rejecting a TRUE H0 (probability = α, the level of significance, typically 1%, 5%, or 10%). Type-II error: failing to reject a FALSE H0 (probability = β).
Two-tail vs. One-tail tests (p.339-344)	Two-tail: H1 uses ≠, rejection regions on both ends (e.g. tyre life ≠ 100K). One-tail: H1 uses < or >, a single rejection region (e.g. mileage < 25 km/l).
z-test / t-test for the mean (p.331)	z-test used when population σ is known (rare in practice); t-test used when σ is unknown (the common case) — uses the t-distribution with df = n−1.
z-test for the proportion (p.353-355)	Tests a claim about a population proportion p using the sample proportion $\underline{p}$ and its standard error, referenced against the Z distribution.

Chapter 10: Two-Sample Tests & One-Way ANOVA *(pp. 357—412)*

Means of two independent populations (p.361-368)	Compares $\mu 1$ vs $\mu 2$ from two separate samples (e.g. prepaid vs. postpaid talk time) — typically a t-test with pooled/unequal variances.
Means of two related (matched-pair) populations (p.375-380)	Compares paired observations on the SAME subjects (e.g. before/after) — a paired t-test on the differences.
Proportions of two independent populations (p.381-385)	Compares p1 vs p2 from two independent samples using a z-test on the difference of proportions.
F-test for two variances (p.386-390)	Tests whether two population variances are equal — ratio of the two sample variances compared against the F-distribution; often a precursor to choosing the correct two-sample t-test (pooled vs. unequal variance).
One-Way ANOVA (p.391-412)	Tests whether 3+ population means are equal: H0: μA = μB = μC ... Compares Variance Among groups (sA^2) to Variance Within groups (sW^2) via the F-test/F-ratio; a large F → reject H0 (means differ). Computable in MS Excel's Data Analysis ANOVA tool.

Chapter 12: Simple Linear Regression *(pp. 413—454)*

Regression equation (p.437-445)	Y = a + b•X (a = intercept, b = slope). Slope (b) computed first from covariance/variance of X, then intercept (a) derived from the means and slope. Determines the best-fit straight line describing how Y changes with X.
Measures of variation / R^2 (p.446-447)	Coefficient of Determination R^2 = 1 − SSE/SST (SSE = Sum of Squares Error/residuals, SST = Total Sum of Squares) = r^2 (square of correlation coefficient). Ranges 0 to 1: R^2=1 → perfect fit (all points on the line); R^2 near 0 → poor fit.
Types of Regression (p.450)	Simple Linear Regression (one predictor, straight-line relationship) vs. other/multiple regression types briefly introduced as extensions.

Chapter 11: Chi-Square Tests *(pp. 455—472)*

Chi-square (χ^2) distribution (p.459)	Positively skewed (not symmetric); values range 0 to +∞; defined by degrees of freedom, df (e.g. df = no. of categories − 1 for goodness-of-fit).
Chi-square Goodness-of-Fit test (p.465-471)	H0: the population/observed distribution matches an expected distribution (e.g. is Poisson-distributed); H1: it does not. Compares observed vs. expected frequencies; large χ^2 → reject H0. Applications include testing whether defect counts follow a Poisson distribution.

Nonparametric Methods (Chapter 14 in text) *(pp. 473—492)*

When to use (p.481)	Used when data don't meet parametric test assumptions (e.g. not Normally distributed) or when data are only Ordinal (ranked) rather than Interval/Ratio.
Spearman Rank Correlation (p.475-480)	Correlation between two variables measured using RANKS rather than raw values — nonparametric counterpart to Pearson's r; used e.g. to compare ranking systems (NIRF rankings).
Sign Test (p.482)	One-sample nonparametric test about the Median (not the Mean); based on counting +/− signs of deviations from the hypothesized median.
Mann-Whitney Rank-Sum Test (p.483)	Two-sample nonparametric test comparing whether the medians of two independent populations are equal — nonparametric counterpart to the two-sample t-test.
Kruskal-Wallis Test (p.484)	Compares ranks across 3 or more independent samples — nonparametric counterpart to One-Way ANOVA.

Financial & Management Accounting — Index for Open-Book Exam - Index

Condensed study index (241 pp.) — organized by topic with page references to the original file for the exam.

1. IFRS / IAS Standards *(pp. 2—20)*

IFRS S1 / S2 (p.2-3)	Sustainability & climate-related financial disclosures, emphasizing transparency, relevance, consistency. S1 (general requirements): covers governance, strategy, risk management, and metrics/targets. S2 (climate): guidelines for disclosing climate-related risks/opportunities; aligns with the TCFD framework (Task Force on Climate-related Financial Disclosures); also covers governance, strategy, risk management, metrics.
IFRS 5 (p.4)	Non-current assets held for sale & discontinued operations — classification, measurement, presentation. Asset must be available for immediate sale in its present condition and the sale must be highly probable.
IFRS 7 (p.5)	Financial Instruments: Disclosures. Requires disclosure of the significance of financial instruments and details of risk exposure — credit, market, and liquidity risk — to encourage transparency of impact.
IFRS 8 (p.6)	Operating Segments. Objective: disclose information to help users evaluate the nature and financial effects of the business activities and economic environments the entity operates in.
IFRS 13 (p.7-10)	Fair Value Measurement. FV = exit price in an orderly transaction (market-based, not entity-specific). Fair Value Hierarchy: Level 1 — quoted prices in active markets for identical assets (most reliable; e.g. listed equity securities). Level 2 — observable inputs other than quoted prices, e.g. observable data for similar assets or quoted prices in inactive markets (e.g. corporate bonds valued via market comparables). Level 3 — unobservable inputs based on management assumptions (least reliable; e.g. private equity investment valuation).
IFRS 15 (p.11-14)	Revenue from Contracts with Customers — 5-Step Model: (1) Identify the contract (approved by parties, enforceable rights/obligations); (2) Identify performance obligations; (3) Determine transaction price (incl. fixed/variable consideration — discounts, rebates, bonuses); (4) Allocate the transaction price to obligations; (5) Recognize revenue as/when obligations are satisfied. Practical applications: Software/SaaS — license fees vs. service revenue (upfront license + updates); Construction — percentage-of-completion method (revenue recognized over time).
IAS 16 (p.15-16)	Property, Plant and Equipment (PP&E) — prescribes accounting treatment: recognition, measurement at cost, depreciation, and derecognition.
IAS 2 (p.17-19)	Inventories — measured at the LOWER of Cost and Net Realizable Value (NRV). Cost = purchase price + conversion costs + other costs. NRV = Selling price − Completion/selling costs. Scope: finished goods, WIP, raw materials (excludes financial instruments, biological assets). Allowed methods: FIFO, Weighted Average Cost, Specific Identification. LIFO is prohibited under IFRS (permitted under US GAAP). Write-downs are reversible under IAS 2 but irreversible once made under US GAAP.
US GAAP vs IFRS — R&D (p.20)	US GAAP: R&D expensed as incurred (except internally developed software). IFRS: Research costs expensed; Development costs capitalized once specific criteria are met — this generally leads to higher reported assets and profits under IFRS than under US GAAP.

2. 12 Red Flags — Financial Statement Analysis *(pp. 21—32)*

Warning signs of earnings manipulation / accounting fraud:

- Transactions boosting profit without real performance — unusual asset sales, debt-for-equity swaps, other one-off items that inflate reported profit and mislead on operational success (e.g. selling a building to book a gain while the core business underperforms).

- Accounts receivable growth > sales growth — may signal relaxed credit terms (risk of defaults) or 'channel stuffing' (pushing excess goods to distributors before quarter-end to record revenue early).
- Inventory build-up > sales growth — finished-goods build-up can signal slowing demand (→ price cuts/write-downs); WIP build-up may reflect expected demand; raw-material build-up may signal procurement inefficiency.
- Earnings—cash flow gap increasing — net income rising while operating cash flow is flat/falling suggests aggressive accrual estimates or revenue recognition (e.g. aggressive % of completion use without matching cash inflow).
- Earnings—taxable income gap increasing — growing divergence between book and taxable income with no tax-law change may indicate aggressive accounting for financial reporting (e.g. underestimating warranty expense for books).
- Complex financing structures — R&D partnerships, Special Purpose Entities (SPEs), or selling receivables with recourse can understate liabilities/overstate assets (classic example: Enron's SPE structures used to hide debt).
- Large asset write-offs — sudden, unexpected impairments suggest delayed recognition of poor performance (e.g. writing off an overvalued acquisition).
- Large 4th-quarter adjustments — significant year-end adjustments following audit suggest interim statements were overly optimistic (e.g. large bad-debt provisions booked only at year-end).
- Auditor-related concerns — qualified opinions or unexplained auditor changes may indicate 'opinion shopping' (e.g. switching auditors after a dispute over revenue recognition).
- Related-party transactions — deals with affiliated entities/insiders may be non-arm's-length, with biased estimates (e.g. inflated sales to a sister company to boost revenue).
- Off-balance-sheet items & contingencies — unexplained growth in guarantees/commitments can conceal debt or risk (e.g. large operating lease commitments not recorded as liabilities).

3. Fundamental / Ratio Analysis *(pp. 33—65)*

Fundamental analysis = Economic analysis + Industry analysis + Company (ratio) analysis. Five ratio families: Liquidity, Profitability, Turnover/Activity, Solvency/Leverage, Valuation.

Liquidity Ratios

Current Ratio	Current Assets / Current Liabilities (ideal ≈ 2:1). CA = anything convertible to cash within a year: cash, bank, ST investments, stock, debtors, o/s income, prepaid exp. CL = payable within a year: ST loans, creditors, bank overdraft, o/s expense, pre-received income.
Liquid / Quick / Acid-Test Ratio	Liquid Assets / CL, where Liquid Assets = CA − Inventory − Prepaid Expenses.
Absolute Liquid Ratio	(Cash in hand + Bank + ST Investments/Marketable Securities) / CL — the strictest liquidity test.

Profitability Ratios

Gross Profit Ratio	Gross Profit / Net Sales × 100. GP = Sales − COGS; COGS = Opening Stock + Net Purchases + Direct Expenses − Closing Stock.
Operating Profit Ratio	EBIT / Sales × 100 (EBIT = Sales − all expenses except interest & tax).
Net Profit Ratio	Profit after Tax / Net Sales × 100.
EPS	(Net Profit after Interest & Tax − Preference Dividend) / Number of Equity Shares.
Return on Equity (ROE)	(Net Profit after Tax − Preference Dividend) / Equity Shareholders' Funds × 100.
ROI / ROCE	EBIT / Capital Employed × 100, where Capital Employed = Debt + Equity Capital.

Solvency (Leverage) Ratios

Debt-Equity Ratio	Debt / Equity.
Debt Capital Ratio	Debt / (Debt + Equity).

Interest Coverage Ratio	EBIT / Interest Paid — how many times over interest obligations are covered by earnings.

Turnover (Activity) Ratios

Debtors Turnover Ratio / DSO	Net Credit Sales / Average Debtors. Avg Collection Period (DSO) = 365 / Debtors Turnover.
Creditors Turnover Ratio / DPO	Net Credit Purchases / Average Payables. Avg Payment Period (DPO) = 365 / Creditors Turnover.
Inventory Turnover Ratio / DIO	COGS / Average Inventory at Cost. Inventory Turnover Days (DIO) = 365 / Inventory Turnover.
Fixed Asset Turnover Ratio	Revenue from Operations / Net Fixed Assets.

Note: turnover ratios can also be expressed per 52 weeks / 12 months instead of 365 days.

Valuation Ratio & DuPont Analysis

P/E Ratio	Market Price per Share / EPS — indicates whether a stock looks expensive or cheap relative to earnings.
DuPont 3-Step Analysis	ROE decomposed into: (1) Profitability — Net Profit Margin; (2) Efficiency — Asset Turnover; (3) Leverage — Equity Multiplier.

Industry benchmarking (from case discussion): banks → high receivables/payables, high leverage, low asset turnover, little inventory. Retailers → high inventory, low GP%, high asset turnover. High-tech/pharma → high R&D, high margins, low debt. Services → high receivables, low inventory, high PP&E. Capital-intensive (airlines, oil & gas, liquor) → high PP&E or inventory aging, often debt-financed.

4. Cash Flow Statement *(pp. 94—103)*

The Balance Sheet shows assets & financing but not the period's changes; the P&L shows performance on an accrual basis but not cash generated. The Cash Flow Statement bridges this — it shows changes in financial position on a cash basis: all cash inflows for a period less cash outflows, showing sources & applications of cash and the net cash in hand at year-end.

Operating Activities	All activities tied to actual business operations — manufacture, sale, and delivery of the product/service the entity deals in; related cash inflows/outflows.
Investing Activities	Actual investments — asset acquisition and money from asset sales; purchase/redemption of investments (e.g. shares of other companies), loans given to others.
Financing Activities	Raising finance via shares, debentures, or loans; dividends paid to shareholders and interest paid on loans are outflows here.

Net cash flow from operating activities can be computed by the Direct Method or the Indirect Method. Layout: Net Profit ± Operating adjustments ± Long-term Investing items ± Financing items = Net Increase/Decrease in Cash; + Cash at beginning of year = Cash at end of year.

5. Cost Classification (Managerial Accounting) *(pp. 120—150)*

Management accounting = classifying, recording, and appropriately allocating expenditure to determine cost of products/services, for use in managerial decision-making.

By Traceability	Direct Costs (Prime Cost) — directly traceable to a product: Direct Material (becomes part of the finished product, tracked per unit), Direct Labour (work clearly traceable to a job/unit), Direct Expenses (e.g. royalties per unit, job-specific tooling — necessary but not part of the product). Prime Cost = DM + DL + DE. Indirect Costs/Overheads — Manufacturing, Administrative, and Selling & Distribution overheads (indirect material + indirect labour + indirect expenses).
By Variability/Behaviour	Fixed Costs — total stays the same regardless of units produced (e.g. rent, factory manager's salary). Variable Costs — cost per unit constant, total changes linearly with output (e.g. direct material/labour are always

	fully variable). Semi-Variable Costs — mixed fixed + variable element. Marginal/Variable cost = change in total cost for a 1-unit change in output.
By Relevance to Decisions	Relevant/Avoidable Costs & Opportunity Costs — differ between decision alternatives, so used in decision-making. Irrelevant/Unavoidable Costs & Sunk Costs — already incurred, unaffected by the decision, so ignored (e.g. money already spent on old software is a sunk cost even if a better version is later purchased).

Worked example logic: a chemical bought for ₹500/unit that has no use is a full sunk cost. If a buyer will only take it after paid conversion work, or disposal costs money, the decision compares only the incremental (relevant) cash flows — the sunk purchase cost never enters the calculation.

6. Inventory Valuation Methods *(pp. 151—154)*

These methods determine the value of inventory on hand, which drives COGS and therefore profit and taxes.

FIFO (First-In, First-Out)	Oldest inventory assumed sold first. In rising prices → lower COGS, higher ending inventory value → higher profit & higher tax. Best for perishables/items with a shelf life.
LIFO (Last-In, First-Out)	Most recently purchased inventory assumed sold first. In rising prices → higher COGS, lower ending inventory → lower profit & lower tax. Best where firms want to reduce tax liability in a rising-price environment. Prohibited under IFRS/IAS 2, permitted under US GAAP.
Weighted Average Cost (WAC)	A single average cost per unit = total cost of goods available / units available; recalculated with every purchase. Produces smoother, less volatile valuations. Best for large quantities of identical/indistinguishable items (e.g. bulk commodities).

7. Budgeting *(pp. 155—160)*

Segregating costs (fixed vs variable, direct vs indirect) underpins planning and preparing budgets for the year ahead.

Cash Budget	Plan estimating cash inflows (sales revenue, loans, capital contributions) and outflows (opex, debt repayment, inventory purchase) over a period, ensuring enough liquidity to meet short-term obligations. Net Cash Flow = Inflows − Outflows. Advantages: predicts cash needs/shortages; manages short-term obligations; guides borrowing/investing decisions.
Master Budget	Comprehensive plan consolidating all individual budgets — Sales Budget, Production Budget, Direct Materials Budget, Direct Labour Budget, Overhead Budget, and Cash Budget — into one overall financial picture. Advantages: holistic planning; cross-department coordination; performance control (actual vs budget); supports decision-making.
Zero-Base Budget (ZBB)	Every expense must be justified from a 'zero base' each period rather than starting from the prior year's figures. Every line item is justified on importance/necessity; resources prioritized toward value-adding activities rather than historical allocation.

8. Cost-Volume-Profit (CVP) & Break-Even Analysis *(pp. 202—236)*

CVP links cost structure, sales price, and volume to profitability — used for pricing, accept/reject of orders below normal price, make-vs-buy, and product-mix decisions under limited resources.

Sales = Profit + Fixed Cost + Variable Cost → S − VC = Contribution = FC + P

Total Contribution	Total Sales − Total Variable Cost
Contribution per unit	Selling Price per unit − Variable Cost per unit
PV Ratio (Contribution Margin %)	Contribution / Sales × 100 (= Cont. p.u. / SP p.u. × 100). Identity: VC% + PV Ratio% = 100%

Break-Even Point (units)	Fixed Cost / Contribution per unit
Break-Even Point (₹ sales)	Fixed Cost / PV Ratio (equivalently, BEP units × SP per unit)
Margin of Safety (₹)	Profit / PV Ratio. Identity: Margin of Safety + BEP Sales = Actual/Total Sales
Margin of Safety (units)	Profit / Contribution per unit
Sales (₹) for a desired profit	(Fixed Cost + Desired Profit) / PV Ratio
Sales (units) for a desired profit	(Fixed Cost + Desired Profit) / Contribution per unit

Worked check (BEP): SP=₹100/u, VC=₹40/u, FC=₹18,000/yr → PV Ratio=(100−40)/100=60%; BEP=18,000/60%=₹30,000 (=300 units). Verify: Sales 30,000 − VC 12,000 = Contribution 18,000 = FC 18,000 → Profit nil.

Make-or-Buy & Special Order Decisions

- Make-or-Buy: Compare marginal (variable) cost of manufacturing vs. outside purchase price. All variable costs are relevant; fixed costs are irrelevant unless the decision itself changes them. Fixed cost per unit should be spread over units actually manufactured, not over capacity. If idle capacity created by buying can be used elsewhere, include that opportunity contribution (or hire-out income) in the comparison — pick whichever alternative has the lowest net cost.
- Special Orders: Accept only if (i) spare capacity exists, (ii) the offered price at least covers the differential/marginal cost of the extra units, and (iii) there's no adverse knock-on effect on regular customers' pricing expectations — a below-normal price to a new/foreign buyer using idle capacity can still be worthwhile even if it wouldn't be offered to a regular local customer.

9. Standard Costing & Variance Analysis *(pp. 237—241)*

Standard cost = predetermined cost based on a technical estimate of materials, labour, and overhead for a given period and prescribed working conditions. Variances between standard and actual help control costs and diagnose the reasons for deviations.

Material Cost Variance (MCV)	Std Cost of Material − Actual Cost of Material = (SQ×SP) − (AQ×AP)
Material Price Variance (MPV)	AQ × (SP − AP) — the price element of MCV
Material Usage Variance (MUV)	SP × (SQ − AQ) — the quantity element of MCV. Check: MPV + MUV = MCV
Labour Cost Variance (LCV)	Std Cost of Labour − Actual Cost of Labour = (ST×SR) − (AT×AR)
Labour Rate Variance (LRV)	AT × (SR − AR) — the rate element of LCV
Labour Efficiency Variance (LEV)	(ST − AT) × SR — the time/efficiency element of LCV. Check: LRV + LEV = LCV

Where AQ = Actual Quantity, AP = Actual Price/unit, SQ = Standard Quantity, SP = Standard Price/unit, AT = Actual Time, AR = Actual Rate, ST = Standard Time, SR = Standard Rate. Critically: SQ/ST is the standard quantity/time allowed for the ACTUAL output achieved (not the original budgeted output) — e.g. if the standard is 5 kg per unit and actual output is 200 units, SQ = 5 × 200 = 1,000 kg.

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Note: "D1 p.#" refers to the page number in the Lectures 1—9 PDF; "D2 p.#" refers to the page number in the Lectures 10—16 PDF. Sixteen lecture topics, each with key sub-topics and page references, are listed in order below.

#	TOPIC / SUB-TOPIC	PAGE	
3	**DIVERSITY IN ORGANIZATIONS**	**D1 p.54—72**	*Examines dimensions of workforce diversity and their link to attitudes and job satisfaction.*
	Levels of Diversity	D1 p.57	Surface-level (biographical) vs. deep-level (values) diversity
	Biographical Characteristics	D1 p.59	Age, gender, race, tenure, disability as diversity variables
	Diversity Management Strategies	D1 p.60	Attracting, developing, retaining diverse talent
	Attitudes & Their Components	D1 p.61—62	Cognitive, affective, behavioral components
	Job Attitudes & Job Satisfaction	D1 p.64—65	Key work attitudes and drivers of satisfaction
	Causes & Outcomes of Job Satisfaction	D1 p.66—67	Pay, work itself, and links to performance/turnover
	Employee Responses to Dissatisfaction	D1 p.68	Exit, voice, loyalty, neglect framework
4	**PERSONALITY AND VALUES**	**D1 p.73—97**	*Explores personality frameworks and value systems that shape workplace behavior.*
	Personality Determinants	D1 p.78	Heredity and environment shape personality
	Myers-Briggs Type Indicator (MBTI)	D1 p.79—81	16-type personality classification tool
	Big Five Model (OCEAN)	D1 p.82—85	Extraversion, agreeableness, conscientiousness, stability, openness
	The Dark Triad	D1 p.86	Machiavellianism, narcissism, psychopathy at work
	Values: Terminal & Instrumental	D1 p.87	End-state goals vs. preferred means/behaviors
	Hofstede's Framework	D1 p.88—89	Cultural dimensions — power distance, individualism, etc.
	Case: Costs of Being Nice	D1 p.91—96	Case on trade-offs of high agreeableness at work
5	**PERCEPTION AND INDIVIDUAL DECISION MAKING**	**D1 p.98—111**	*Covers how perception biases judgment and shapes individual decision-making.*
	What Is Perception & Influencing Factors	D1 p.101—102	Perceiver, target, situation shape perception
	Attribution Theory	D1 p.103—104	Internal vs. external causes of behavior
	Common Shortcuts in Judging Others	D1 p.105	Stereotyping, halo effect, selective perception
	Decision-Making Types & Rationality Model	D1 p.106—108	Programmed/non-programmed decisions; 6-step rational model
	Common Biases & Errors in DM	D1 p.109	Overconfidence, anchoring, availability biases
	Three Ethical Decision Criteria	D1 p.110	Utilitarianism, rights, justice
6	**EMOTIONS AND MOODS**	**D1 p.112—136**	*Distinguishes emotions from moods and examines their sources, management, and organizational impact.*

#	TOPIC / SUB-TOPIC	PAGE	
	Emotions vs. Moods Defined	D1 p.115	Emotions intense & targeted; moods diffuse
	Six Universal & Moral Emotions	D1 p.116—117	Anger, fear, sadness, happiness, disgust, surprise
	Basic Moods: Positive/Negative Affect	D1 p.118	Two primary mood dimensions
	Sources of Emotions & Moods	D1 p.119—120	Personality, day/time, weather, stress, sleep, exercise
	Emotional Labor	D1 p.121—122	Managing displayed emotions to meet job demands
	Affective Events Theory	D1 p.123	Workplace events trigger emotions affecting performance
	Emotional Intelligence	D1 p.124—125	Ability to perceive & manage own/others' emotions
	OB Applications & Implications for Managers	D1 p.126—128	Uses in selection, motivation, decisions, leadership
	Case: Crying at Work	D1 p.129—135	Case on emotional display norms in the workplace
7	**MOTIVATION CONCEPTS**	**D1 p.137—167**	*Reviews early and contemporary motivation theories explaining what drives employee effort.*
	Maslow's Hierarchy of Needs	D1 p.143	Five-level needs hierarchy
	Herzberg's Motivation-Hygiene Theory	D1 p.145	Hygiene factors prevent dissatisfaction; motivators drive satisfaction
	Goal-Setting Theory	D1 p.151	Specific, difficult goals improve performance
	Equity Theory	D1 p.153—155	Employees compare input—outcome ratios to others
	Expectancy Theory	D1 p.157—159	Effort—performance—reward linkages drive motivation
	Case Study & Questions	D1 p.160—166	Applied motivation-theory case scenarios
8	**MOTIVATION: FROM CONCEPTS TO APPLICATIONS**	**D1 p.168—186**	*Translates motivation theory into practical job design and reward-system applications.*
	Job Characteristics Model (JCM)	D1 p.171	Skill variety, task identity/significance, autonomy, feedback
	Job Redesign, Rotation, Enrichment	D1 p.172—174	Restructuring jobs to boost motivation
	Alternative Work Arrangements	D1 p.175—177	Flextime, job sharing, telecommuting
	Employee Involvement	D1 p.178	Participative management, representation, quality circles
	Using Rewards & Benefits to Motivate	D1 p.179—180	Variable pay, skill-based pay, flexible benefits
	Case: Carrot to the Stick	D1 p.181—185	Case on reward-system design trade-offs

#	TOPIC / SUB-TOPIC	PAGE	
	Leadership vs. Management	D2 p.36—37	Coping with change vs. coping with complexity
	Contingency Theories of Leadership	D2 p.39—42	Leader effectiveness depends on situational fit
	House's Path-Goal Model	D2 p.43	Leader clarifies followers' path to goals
	Charismatic Leadership	D2 p.45—46	Vision-based influence; born-or-made debate
	Transformational vs. Transactional Leaders	D2 p.47	Inspiring change vs. exchange-based leadership
	Full Range of Leadership Model	D2 p.48	Continuum from laissez-faire to transformational
	Leadership Point of View; Role of a Leader	D2 p.51—52	Follower perspective and the leader's core role
12	**ORGANIZATIONAL STRUCTURE**	**D2 p.54—78**	*Details the six structural elements and common organizational design types.*
	Six Elements of Organization's Structure	D2 p.57—59	Specialization, departmentalization, chain of command, span, centralization, formalization
	Departmentalization	D2 p.60	Grouping jobs by function, product, geography, etc.
	Chain of Command & Span of Control	D2 p.61—64	Authority lines and number of direct reports
	Centralization/Decentralization; Formalization	D2 p.65—66	Decision-authority location; rule standardization
	Common Designs: Simple, Bureaucracy, Matrix	D2 p.68—70	Structural trade-offs across design types
	Virtual, Team, Circular Structures	D2 p.71—73	Contemporary flexible organizational forms
	Strategy and Structure	D2 p.75	Structure follows strategy (Chandler)
13	**ORGANIZATIONAL CULTURE & CHANGE MANAGEMENT**	**D2 p.79—113**	*Explains how culture forms and persists, then covers models for managing organizational change.*
	Organizational Culture Defined	D2 p.82—83	Shared meaning system distinguishing organizations
	Strong vs. Weak Cultures	D2 p.84	Degree to which shared values guide behavior
	How Culture Begins & Is Kept Alive	D2 p.85—86	Founder influence; selection, socialization, top management
	Socialization Model	D2 p.87	Pre-arrival, encounter, metamorphosis stages
	Forces & Types of Organizational Change	D2 p.93—94	External/internal drivers; planned vs. unplanned
	Resistance to Change	D2 p.96	Individual and organizational sources of resistance
	Kurt Lewin's Model	D2 p.98	Unfreezing, changing, refreezing
	Action Research & Organizational Development	D2 p.99—100	Data-driven change; humanistic OD techniques

#	TOPIC / SUB-TOPIC	PAGE	
	Reducing Resistance to Change	D2 p.104	Communication, participation, negotiation tactics
	Case Studies 1—3	D2 p.105—113	Applied change-management scenarios
14	**RECRUITMENT & SELECTION**	**D2 p.114—142**	*Covers job analysis, the recruitment process, and selection methods used to hire employees.*
	Job Analysis	D2 p.120—121	Determining job duties, skills, requirements
	Purpose, Importance & Factors in Recruitment	D2 p.122—123	Sourcing candidates; internal/external influences
	Recruitment Process	D2 p.124—127	Stages from requisition to candidate pool
	Philosophies & Alternatives to Recruitment	D2 p.128—129	Outsourcing, overtime, temp staffing options
	Selection: Outcomes & Process	D2 p.131—132	Screening steps from application to offer
	Types & Problems of Interviews	D2 p.133—134	Structured vs. unstructured; bias pitfalls
	Selection Challenges & Effectiveness	D2 p.135—139	Reliability, validity, legal considerations
	Work-Life Balance	D2 p.140—142	Balancing work demands with personal life in hiring
15	**COMPENSATION MANAGEMENT**	**D2 p.143—174**	*Explores compensation philosophy, theories, and the process of designing pay structures.*
	Compensation Philosophy & Components	D2 p.147—148	Direct and indirect pay elements
	Theories of Compensation	D2 p.149—152	Expectancy and equity theory applied to pay
	Consequences of Pay Dissatisfaction	D2 p.153	Turnover, grievances, reduced performance
	Motivation—Performance Model	D2 p.154	Links pay to performance outcomes
	Factors Influencing Remuneration	D2 p.156—157	Internal equity, market rates, business strategy
	Devising a Compensation Plan	D2 p.158	Steps to design pay structures
	Challenges of Compensation	D2 p.160	Pay equity, transparency, cost control
	Case: Pay Equity Dilemma at TechNova	D2 p.162—174	Case on gender/role-based pay disparity
16	**PERFORMANCE APPRAISAL**	**D2 p.175—207**	*Details the performance appraisal process, rating problems, and modern appraisal methods.*
	PA Process & Objectives	D2 p.180—181	Steps and purposes of appraising performance
	Establish Job Expectations & Design Program	D2 p.183—184	Setting standards before appraisal design
	Rating Problems & Solutions	D2 p.185—186	Halo, leniency, central tendency errors & fixes
	What & When to Evaluate	D2 p.187—188	Traits, behaviors, results; timing choices
	Methods: Rating Scales, Checklist, BARS	D2 p.190—195	Graphic scales through behaviorally anchored scales

#	TOPIC / SUB-TOPIC	PAGE	
	MBO, Psychological Appraisals, Assessment Centres	D2 p.198—200	Goal-based, psych-based, simulation-based methods
	360° and 720° Appraisal	D2 p.201	Multi-rater and multi-cycle feedback approaches
	Appraisal Interview & Performance Management	D2 p.204—205	Feedback delivery; broader PM system
	Emerging Trends in Performance Management	D2 p.206	Continuous feedback, agile appraisal shifts

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Page numbers refer to the slide/page number in the course PDF (BITS WILP Study Material).

Topic / Sub-topic		Page(s)
1. Introduction to Marketing & Core Concepts *(pp. 7—46)*		
Defining Marketing for New Realities	Chapter opener introducing marketing's evolving role in the firm.	7
Value & Scope of Marketing	Marketing works with other functions to build brand value, profit and customer loyalty.	8—12
Case: Tata Curvv Marketing	Illustrates product design, institutional partnerships and launch process for an EV.	13—14
Four Concepts of Marketing	Traces evolution from Production/Product/Selling concepts to today's customer-centric Marketing concept.	15—16
Scope: what is marketed / who is the marketer	Marketing covers goods, services, experiences, ideas etc.; marketer seeks a response from a prospect.	17, 20—21
Marketing & Selling distinguished	Selling is only one activity within the broader marketing function.	19
Needs, Wants & Demands; Maslow's Hierarchy; Demand States	Differentiates needs/wants/demands and links them to Maslow's hierarchy and demand states.	22—26
What is a Market; Key Customer Markets	Defines "market" and maps money/resource flow among producers, government and consumers.	27—29
Marketing Process (overview)	Marketing = choosing target markets + creating, delivering, communicating superior value.	31
Offerings, Brands, Value & Satisfaction	Covers STP, market offers, and how brands create customer value and satisfaction.	32—34
Value Proposition — developing (5 steps)	Functional, economic, psychological, social & hedonic value, illustrated via BMW/VW.	35—41
Marketing Channels & Supply Chain	Distinguishes distribution/service/communication channels and the raw-material-to-consumer chain.	42—43
Promotion & Classification of Media (POEM)	Impressions vs engagement; classifies media into Paid, Owned and Earned.	44—45
Competition (core concept)	Competition = actual/potential rivals and substitutes, illustrated with automobiles.	46
2. Marketing Strategy & Marketing Mix *(pp. 47—61)*		
Marketing Strategy / Go-to-Market	Marketing mix is the tactical bridge between strategy and the target market.	47
Marketing Mix & Developing Mixes for Target Markets	Marketing mix = set of levers combined to elicit a target-market response.	48—49

AIDA Model & Hierarchy of Effects	Attention-Interest-Desire-Action and hierarchy-of-effects models, with examples.	362—366
Developing Effective Communications (8 steps)	Eight-step process from identifying the target audience to choosing message/channel.	367—370
Message & Creative Strategy; Message Source	Rational, sensory, social and ego message appeals; message-source credibility.	371—375
Establishing the Communications Budget (Objective-and-Task method)	Affordable, percentage-of-sales, competitive-parity and objective-and-task budgeting.	376—377
Selecting the Communications Mix & its Characteristics	Reviews strengths of each channel — advertising, sales promotion, PR, etc.	378—383
Setting the Marketing Communications Mix	Product-market type and product life-cycle stage shape the ideal communications mix.	384—385
Managing Integrated Marketing Communications (IMC) — steps	IMC = coordinating all brand contacts to be consistent and relevant over time.	386—388
Digital Marketing (terms, paid ads & keywords)	SEO/GEO, SEM keywords, and digital metrics — impressions, clicks, conversions.	389—393
9. Distribution & Marketing Channels (Place) *(pp. 394—451)*		
Learning Objectives	Frames channel systems, design, management and e-commerce/conflict issues.	396—397
The 'Invisible' P of Marketing — Place; Marketing Channels	Availability (Place) is the "invisible" but crucial P; introduces intermediaries.	398—400
Channel Functions	Channels bridge spatial/temporal gaps, break bulk and provide assortment.	401—402
Types of Intermediaries	Compares distributors, dealers etc. by stocking, ownership and affiliation.	403
Marketing Channels & Value Network; Multi-channel Marketing	Multi-channel marketing and value networks, illustrated by Eureka Forbes.	404—411
Role of Marketing Channels	Channel communication, forward/backward flows and channel-manager relationships.	412—415
Channel Design Steps & Decisions	How firms (Dell, Samsung, LG) design channel structures to reach customers.	416—420
Identifying Major Channel Alternatives (incl. Pharma case)	Channel-alternative types, intermediary counts and responsibilities.	421—425
Channel Control & Adaptability	Trade-off between control (owned channels) and flexibility.	426
Break-Even: Own Sales Force vs Sales Agency	Fixed-cost sales force vs commission-based agency, via break-even volume.	427, 436

Market Expansion Decision (Pricing + Channel, BEP)	Applies break-even analysis to a channel/pricing expansion case (gym example).	428—435
Channel Design Case: Bosch Auto Spare Parts	Designing a spare-parts channel based on customer needs — lot size, delivery time.	437—440
Pricing in FMCG & Consumer Goods Sector	Compares MRP, margins and channel economics across FMCG vs durable goods.	441
Channel Management Decisions	Selecting, training, motivating and evaluating channel members.	442—444
Training & Motivating Channel Members	Five bases of channel power: coercive, reward, legitimate, expert, referent.	445—446
E-Commerce & M-Commerce	Pure-click vs brick-and-click models; mobile issues like geofencing & privacy.	447—448
Conflict, Cooperation, and Competition	Causes of channel conflict and the need for channel coordination.	449—451